

DOCE Assignment – 2

Mallidi Akhil Reddy

2024TM93056

RASA Code - <https://github.com/itzzmeakhi/tripper-bot-be>

FE Web App code - <https://github.com/itzzmeakhi/tripperbot-web>

Tripper Bot – A Travel Assistant

- ▶ Design of Conversational Experiences – Assignment 2

- ▶ Mallidi Akhil Reddy
- ▶ 2024TM93056

Introduction

- ▶ Tripper Bot helps users plan trips by providing:
 - ▶ - Destinations
 - ▶ - Hotels
 - ▶ - Restaurants
 - ▶ - Personalized itineraries
- ▶ Its built from scratch using Rasa framework with custom actions, slots, NLU, and stories.

Bot Persona

- ▶ Name: Tripper Bot
- ▶ Environment: Web travel app
- ▶ Target Audience: Travelers seeking guidance
- ▶ Purpose: Simplify trip planning
- ▶ Core Values: Personalization, reliability, helpfulness
- ▶ Personality Traits: Friendly, knowledgeable, concise

Conversation Flow

User greets → Bot shows menu →
User selects service

Bot asks relevant questions → Bot
provides recommendations

Multi-turn conversation with slot
filling

Handles deterministic responses for
greeting, goodbye, bot challenge

NLU & Intents

- ▶ Key intents:
- ▶ greet, goodbye, find_destination, search_hotels, find_restaurants, plan_itinerary, bot_challenge
- ▶ Example utterances:
- ▶ greet: 'Hi', 'Hello there'
- ▶ find_destination: 'Suggest a place to visit'
- ▶ Entities: destination_type, budget_level, duration, location, cuisine_type

Custom Actions

- ▶ action_recommend_destinations: Filters by type & budget
- ▶ action_recommend_hotels: Filters by location & budget
- ▶ action_recommend_restaurants: Filters by location & cuisine
- ▶ action_create_itinerary: Generates day-wise itinerary
- ▶ Includes highlights, budget, season, amenities, ratings

Stories & Rules

- ▶ Stories: Multi-turn conversation sequences
- ▶ Rules: Deterministic handling of greetings, goodbye, bot challenges
- ▶ Ensures smooth conversation flow and fallback handling

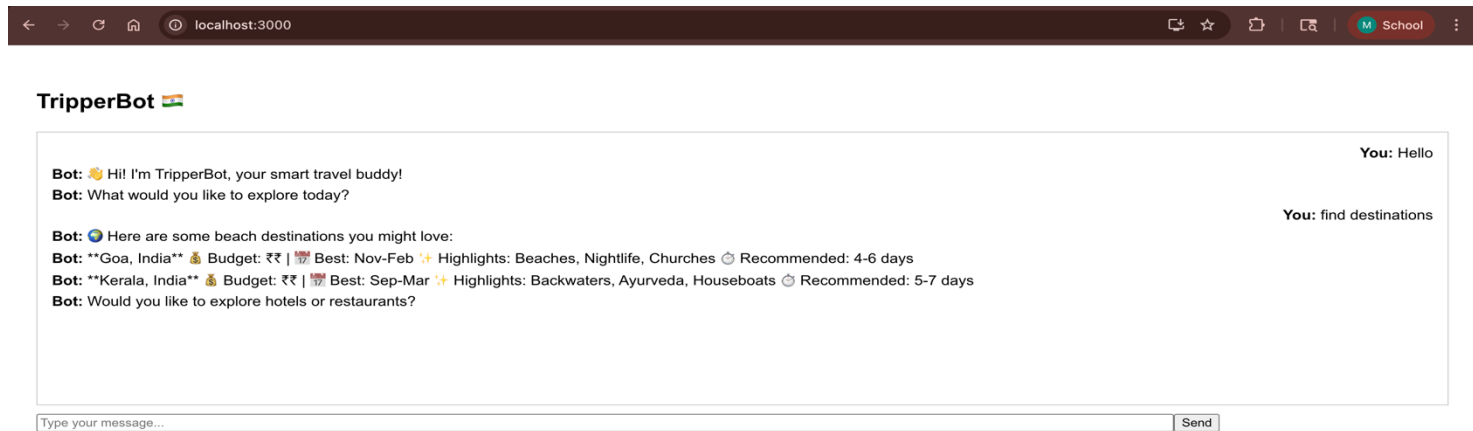
Screenshots / Demo

► RASA Shell

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🌍 Here are some beach destinations you might love:
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📅 Budget: ₹₹ | 📅 Best: Nov-Feb
🌟 Highlights: Beaches, Nightlife, Churches
🕒 Recommended: 4-6 days
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📅 Budget: ₹₹ | 📅 Best: Sep-Mar
🌟 Highlights: Backwaters, Ayurveda, Houseboats
🕒 Recommended: 5-7 days
? Would you like to explore hotels or restaurants? Type out your own message...
Your input -> Show me luxury hotels
🏨 Top hotels in Goa:
Taj Exotica Resort & Spa - ★ 4.9/5 - ₹18,000/night (Luxury)
The Leela Goa - ★ 4.8/5 - ₹21,000/night (Luxury)
Marina Bay Beach Resort - ★ 4.2/5 - ₹5,500/night (Budget)
Your input -> 
```

Screenshots / Demo

► RASA Deployed Web App



Deployment



Platform: Web interface (Rasa UI or integrated client)



Session: 60 mins, slots persist



Future integrations: Weather API, live hotel availability, multi-language support

Summary / Conclusion



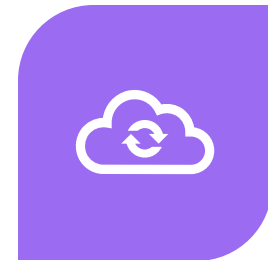
PERSONALIZED MULTI-
SERVICE TRAVEL
ASSISTANT



DEMONSTRATES NLU,
STORIES, RULES, SLOTS,
AND CUSTOM ACTIONS



USER-FRIENDLY WITH
BUTTONS & FREE-TEXT
INPUT



READY FOR REAL-
WORLD DEPLOYMENT
AND ENHANCEMENTS

Thank you

Tripper Bot – A Travel Assistant

MALLIDI AKHIL REDDY

2024TM93056

Introduction

Tripper Bot is a travel assistant chatbot built using Rasa framework. Its primary goal is to help users plan trips by providing personalized destination suggestions, hotel recommendations, restaurant options, and detailed itineraries. Unlike generic chatbots, Tripper Bot uses custom actions, slot filling, and multi-turn conversations to deliver a smooth, contextual, and user-friendly travel planning experience.

Bot Persona

- Name: Tripper Bot
- Environment: Travel website / Web application
- Target Audience: Travelers of all ages looking for guided travel suggestions
- Purpose: Simplify trip planning by providing recommendations based on destination type, budget, duration, and cuisine preferences
- Runtime Variations: Greeting varies with time of day (morning/evening)
- Recommendations Adapt to user-provided preferences and session slots
- Service Branding: Friendly, informative, trustworthy, and travel-themed
- Core Values: Personalization, reliability, efficiency, user satisfaction
- Personality Traits: Friendly, knowledgeable, concise, engaging, professional

Conversation Design

3.1 Key Intents

- greet – User greets the bot
- goodbye – User ends the conversation
- bot_challenge – User asks if the bot is a human
- find_destination – User wants destination suggestions
- search_hotels – User wants hotel recommendations

- find_restaurants – User wants restaurant suggestions
- plan_itinerary – User wants a day-wise travel plan
- provide_destination_type – User specifies type of trip (beach, mountain, city, cultural)
- provide_budget – User specifies budget level (budget, moderate, luxury)
- provide_duration – User specifies trip duration
- provide_location – User specifies city or location
- provide_cuisine – User specifies type of cuisine
- ask_weather – User asks about weather

3.2 Entities and Slots

- **Entities:** destination_type, budget_level, duration, location, cuisine_type, hotel_budget
- **Slots:** These entities are captured in slots to maintain context over multi-turn conversations.

Example:

- destination_type → beach, mountain, city, cultural
- hotel_budget → budget, midrange, luxury

3.3 Stories & Rules

- **Stories:** Define multi-turn conversation flows (e.g., greeting → menu → find destination → ask budget → provide recommendations)
- **Rules:** Deterministic handling for greetings, goodbye, and bot challenge to ensure consistent responses

NLU Training

Intent Examples:

greet → “Hi”, “Hello there”, “Good morning”

find_destination → “I want to travel”, “Suggest a place to visit”

search_hotels → “Find hotels”, “Where can I stay?”

provide_destination_type → “I want a beach vacation”

Entity Recognition: Ensures that user inputs are parsed accurately to capture preferences like location, budget, or cuisine.

Training Approach: Multiple examples per intent improve model accuracy and reduce misclassification.

Custom Actions

action_recommend_destinations: Recommends destinations filtered by destination_type and budget_level

action_recommend_hotels: Recommends hotels based on location and optional hotel_budget

action_recommend_restaurants: Recommends restaurants based on location and cuisine_type

action_create_itinerary: Generates day-wise travel plans based on destination_type, budget_level, and duration

Features: Each recommendation includes highlights, budget, season, amenities, ratings, and special features

Weather API Integration

Purpose:

Enable users to check weather details for any destination before planning their trip.

Implementation:

- **Intent:** ask_weather
- **Entity:** location
- **Action File:** actions.py

Deployment

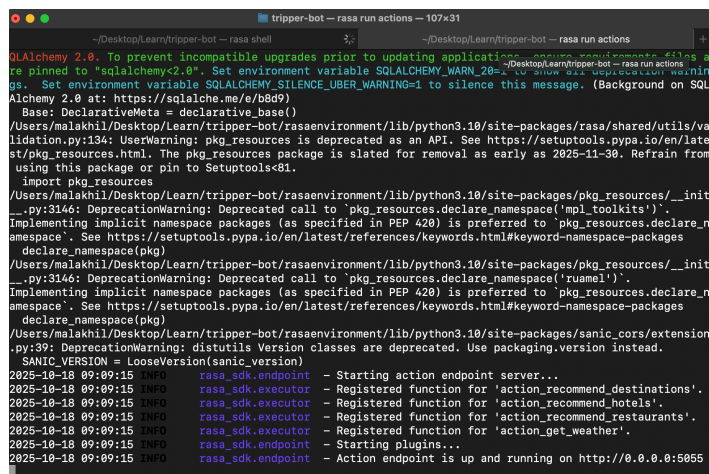
Platform: Web application (Rasa UI or integrated web client)

Next Steps / Integrations:

- Integrate live APIs for weather, flights, or hotel availability
- Expand destination, hotel, and restaurant datasets
- Support multi-language conversations

Screenshots

RASA Actions



```
tripper-bot -- rasa run actions -- 107x31
~/Desktop/Learn/tripper-bot -- rasa shell
~/Desktop/Learn/tripper-bot -- rasa run actions

SQLAlchemy 2.0. To prevent incompatible upgrades prior to updating applications, ensure your SQLAlchemy
is pinned to "sqlalchemy<2.0". Set environment variable SQLAlchemy_WARN_20=1 to show this message. (Background on SQL
Alchemy 2.0 at: https://sqlalche.me/e/b8d9)
Base: DeclarativeMeta = declarative_base()
/Users/malakhil/Desktop/Learn/tripper-bot/rasaenvironment/lib/python3.10/site-packages/rasa/shared/utils/val
idation.py:134: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/late
st/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from
using this package or pin to Setuptools<81.
  import pkg_resources
/Users/malakhil/Desktop/Learn/tripper-bot/rasaenvironment/lib/python3.10/site-packages/pkg_resources/_init
_.py:3146: DeprecationWarning: Deprecated call to 'pkg_resources.declare_namespace('mpl_toolkits')'.
Implementing implicit namespace packages (as specified in PEP 420) is preferred to 'pkg_resources.declare_n
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/Users/malakhil/Desktop/Learn/tripper-bot/rasaenvironment/lib/python3.10/site-packages/pkg_resources/_init
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/Users/malakhil/Desktop/Learn/tripper-bot/rasaenvironment/lib/python3.10/site-packages/sanic_cors/extension
.py:39: DeprecationWarning: distutils Version classes are deprecated. Use packaging.version instead.
  SANIC_VERSION = LooseVersion(sanic_version)
2025-10-18 09:09:15 INFO rasa_sdk.endpoint - Starting action endpoint server...
2025-10-18 09:09:15 INFO rasa_sdk.executor - Registered function for 'action_recommend_destinations'.
2025-10-18 09:09:15 INFO rasa_sdk.executor - Registered function for 'action_recommend_hotels'.
2025-10-18 09:09:15 INFO rasa_sdk.executor - Registered function for 'action_recommend_restaurants'.
2025-10-18 09:09:15 INFO rasa_sdk.executor - Registered function for 'action_get_weather'.
2025-10-18 09:09:15 INFO rasa_sdk.endpoint - Starting plugins...
2025-10-18 09:09:15 INFO rasa_sdk.endpoint - Action endpoint is up and running on http://0.0.0.0:5065
```

RASA Shell

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from jax import xla_computation as _xla_computation
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2025-10-18 09:09:14 INFO      root - Connecting to channel 'cmdline' which was specified by the '--connecto
r' argument. Any other channels will be ignored. To connect to all given channels, omit the '--connector' a
rgument.
2025-10-18 09:09:14 INFO      root - Starting Rasa server on http://0.0.0.0:5005
2025-10-18 09:09:14 INFO      rasa.core.processor - Loading model models/20251018-090801-clear-opacity.tar.
gz...
2025-10-18 09:09:15 WARNING  abs1 - At this time, the v2.11+ optimizer 'tf.keras.optimizers.Adam' runs slo
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RASA Interactive

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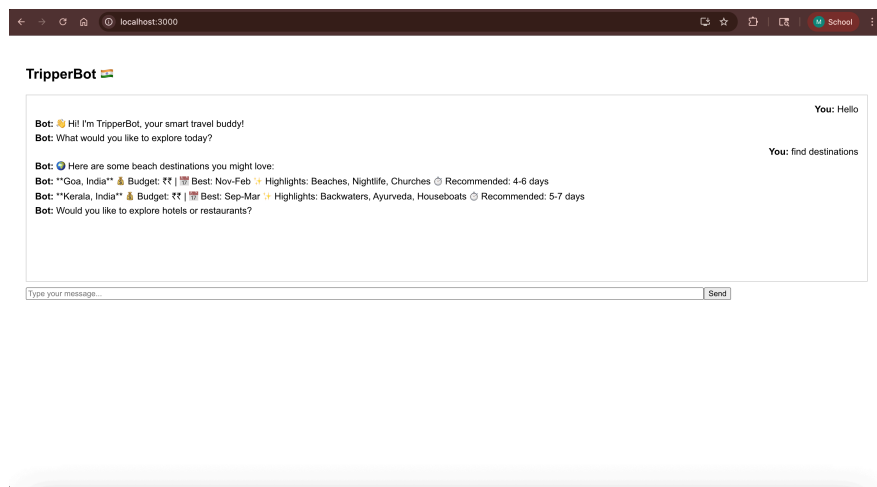
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/Users/malakhi/Desktop/Learn/tripper-bot/rasaenvironment/lib/python3.10/site-packages/tensorflow/lite/pytho
on/util.py:52: DeprecationWarning: jax.xla_computation is deprecated. Please use the AOT APIs.
    from jax import xla_computation as _xla_computation
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2025-10-18 09:14:54 INFO root - Rasa server is up and running.
Processed story blocks: 100% [████████████████████████████████████████] 5/5 [00:00:00:00, 9650.95it/s, # trackers=1]
Processed rules: 100% [████████████████████████████████████████] 5/5 [00:00:00:00, 10205.12it/s, # trackers=1]

Bot loaded. Visualisation at http://localhost:5006/visualization.html .
Type a message and press enter (press 'Ctrl-c' to exit).
? Your input →
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Processed rules: 100%[██████████████████████████████████████████████████████████] 5/5 [00:00:00:00, 10205.12it/s, # trackers=1]  
  
Bot loaded. Visualisation at http://localhost:5006/visualization.html .  
Type a message and press enter (press 'Ctrl-c' to exit).  
? Your input ->
```

RASA Deployed Web App



Conclusion

Tripper Bot is a fully functional travel chatbot built from scratch using Rasa, demonstrating:

- Accurate NLU with intent and entity recognition
- Multi-turn stories for personalized conversation flow
- Custom actions providing dynamic recommendations
- Interactive experience with buttons and fallback handling

The bot provides a user-friendly, engaging, and informative travel planning experience suitable for deployment on web platforms.